

Topic C4

# Electrochemistry

**Concept guide · Worked examples · Cambridge-style practice · Fully explained answers**

## What this guide teaches

A complete explanation of the syllabus concepts, chemical reasoning, worked applications, common misconceptions and original exam-style practice for this topic.

## Syllabus map

|      |                                         |
|------|-----------------------------------------|
| C4.1 | Electrolysis                            |
| C4.2 | Electrolysis of molten compounds        |
| C4.3 | Electrolysis of aqueous solutions       |
| C4.4 | Electroplating and industrial processes |
| C4.5 | Hydrogen-oxygen fuel cells              |

How to study: retrieve definitions and equations, cover the worked examples, then complete the practice paper without using the notes.

# 1. Electrolysis and electrodes

Electrolysis is the decomposition of an ionic compound, molten or in aqueous solution, by electricity. The electrolyte contains mobile ions. The cathode is negative and attracts cations; reduction (gain of electrons) occurs there. The anode is positive and attracts anions; oxidation (loss of electrons) occurs there.

cathode: reduction    anode: oxidation

## Examiner insight

Identify the ions present before predicting products. Electrons move in the external circuit; ions move through the electrolyte.

# 2. Molten ionic compounds

A molten compound contains only its own ions. At the cathode the metal cation gains electrons; at the anode the non-metal anion loses electrons. For molten lead(II) bromide,  $\text{Pb}^{2+}$  forms lead and  $\text{Br}^-$  forms bromine.



## Examiner insight

A solid ionic compound cannot be electrolysed because its ions are not free to move.

# 3. Aqueous electrolysis

Water adds  $\text{H}^+$  and  $\text{OH}^-$  to the possible ions. At an inert cathode, hydrogen forms instead of a metal more reactive than hydrogen; less reactive metal ions are deposited. At an inert anode, concentrated halide ions form halogen; otherwise oxygen commonly forms from hydroxide ions.

## Worked example

Electrolysis of aqueous  $\text{CuSO}_4$  with inert electrodes produces copper at the cathode and oxygen at the anode.

## Examiner insight

Electrode material matters: a copper anode can dissolve and replace  $\text{Cu}^{2+}$  ions removed at the cathode.

## 4. Electroplating and extraction

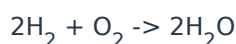
Electroplating coats an object with a metal for appearance or corrosion resistance. The object is the cathode, the coating metal is the anode and the electrolyte contains ions of that metal. Aluminium is extracted from molten aluminium oxide because carbon cannot reduce it; electrolysis is expensive due to electrical energy use.

### Examiner insight

For electroplating, state the electrode roles and the ion in solution, not only the purpose.

## 5. Hydrogen-oxygen fuel cells

A fuel cell converts chemical energy directly to electrical energy while reactants are supplied. Hydrogen combines with oxygen and the only chemical product is water. Advantages include no carbon dioxide at point of use and continuous operation; limitations include hydrogen production, storage, transport and flammability.



### Examiner insight

A fuel cell is not “pollution-free” if the hydrogen was made using fossil fuels; discuss the full energy pathway.

## Exam technique

### High-value habits

Use precise chemical vocabulary, balance equations, show every calculation step, include units and state symbols where required, distinguish observations from conclusions, and answer the command word directly.

## Retrieval checklist

- Recall all definitions, tests and equations without notes.
- Explain one unfamiliar observation using particles, bonding or collision theory.
- Check charge, atoms, units and significant figures in every calculation.
- Complete one timed set and mark it only after finishing.

# Practice Paper

Topic C4

**Electrochemistry****Instructions**

Answer all questions. Select one option for multiple choice. Show working in open spaces and write explanations on the lines. Individual parts carry 1-3 marks.

Candidate name:

Class / date:

**Total marks: 50**

Suggested time: 60 minutes.

## Section A · Multiple choice

Choose the best answer for each question.

1. Where does reduction occur during electrolysis? [1]
- A anode
  - B cathode
  - C electrolyte surface
  - D power supply only
2. Why does solid sodium chloride not conduct? [1]
- A It has no ions.
  - B Its ions are fixed.
  - C Its electrons are delocalised.
  - D Sodium is absent.
3. What forms at the cathode during electrolysis of molten  $\text{PbBr}_2$ ? [1]
- A bromine
  - B oxygen
  - C lead
  - D hydrogen
4. Which half-equation is oxidation? [1]
- A  $\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$
  - B  $2\text{Cl}^- \rightarrow \text{Cl}_2 + 2\text{e}^-$
  - C  $\text{H}^+ + \text{e}^- \rightarrow \text{H}$
  - D  $\text{Ag}^+ + \text{e}^- \rightarrow \text{Ag}$
5. Aqueous copper(II) sulfate with inert electrodes produces at the cathode [1]
- A copper
  - B hydrogen
  - C oxygen
  - D sulfur
6. Concentrated aqueous sodium chloride produces at the anode [1]
- A sodium
  - B hydrogen
  - C chlorine
  - D nitrogen
7. In silver-plating a spoon, the spoon is the [1]
- A positive anode
  - B negative cathode
  - C electrolyte
  - D power supply

8. Why is aluminium extracted by electrolysis? [1]
- A It is less reactive than carbon.
  - B It occurs native.
  - C It is more reactive than carbon.
  - D Its oxide is molecular.
9. What is the product of a hydrogen-oxygen fuel cell? [1]
- A carbon dioxide
  - B water
  - C methane
  - D ammonia
10. Which is a limitation of hydrogen fuel? [1]
- A It always makes soot.
  - B Storage is difficult.
  - C It contains sulfur.
  - D It cannot release energy.

## Section B · Structured questions

11 Molten lead(II) bromide is electrolysed using inert electrodes.

(a) Explain why the molten compound conducts. [2]

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(b) Name the product at each electrode. [2]

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(c) Write the cathode half-equation. [2]

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(d) Write the anode half-equation. [2]

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**[Total: 8]**

12 Dilute aqueous sodium sulfate is electrolysed with inert electrodes.

(a) List the four ions present. [2]

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(b) Predict the cathode product and explain. [2]

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(c) Predict the anode product and explain. [2]

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(d) State the overall effect on water. [2]

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**[Total: 8]**

13 A steel key is electroplated with nickel.

(a) State two reasons for electroplating. [2]

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(b) Identify the electrode connected to the negative terminal. [1]

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(c) State a suitable anode and electrolyte. [2]

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(d) Describe the movement and change of nickel ions. [3]

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**[Total: 8]**

14 Copper(II) sulfate solution is electrolysed using copper electrodes.

(a) Write the cathode half-equation. [2]

(b) Write the anode half-equation. [2]



(c) State the change in mass of each electrode. [2]

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(d) Explain why blue colour remains approximately constant. [2]

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**[Total: 8]**

15 A car uses a hydrogen-oxygen fuel cell.

(a) Write the overall equation. [2]

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(b) State two advantages at point of use. [2]

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(c) State two difficulties with hydrogen. [2]

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(d) Explain why environmental impact depends on hydrogen production. [2]

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**[Total: 8]**

# Worked solutions

**Remove or fold back this section while completing the practice paper.**

## Marking routine

Mark each part separately. Award method and chemistry marks where shown, then identify the first point at which an answer becomes incorrect.

## Solutions · Multiple choice

- 1 B. Cations gain electrons at the cathode.
- 2 B. The ions cannot carry charge through the solid.
- 3 C.  $\text{Pb}^{2+}$  ions gain electrons.
- 4 B. Electrons appear as products, so they are lost.
- 5 A. Copper is less reactive than hydrogen and is deposited.
- 6 C. Chloride ions are oxidised to chlorine.
- 7 B.  $\text{Ag}^{+}$  ions gain electrons on the cathode object.
- 8 C. Carbon cannot reduce aluminium oxide.
- 9 B. Hydrogen and oxygen combine to form water.
- 10 B. Low density and flammability complicate storage and transport.

## Solutions · Structured questions

11. 11(a) Ions are mobile and carry charge [2]. (b) Lead at cathode; bromine at anode [2]. (c)  $\text{Pb}^{2+} + 2\text{e}^- \rightarrow \text{Pb}$  [2]. (d)  $2\text{Br}^- \rightarrow \text{Br}_2 + 2\text{e}^-$  [2].

12. 12(a)  $\text{Na}^+$ ,  $\text{SO}_4^{2-}$ ,  $\text{H}^+$ ,  $\text{OH}^-$  [2]. (b) Hydrogen; Na is more reactive than hydrogen [2]. (c) Oxygen; sulfate remains and  $\text{OH}^-$  is discharged [2]. (d) Water decomposes into hydrogen and oxygen [2].

13. 13(a) Appearance and corrosion/wear resistance [2]. (b) Steel key [1]. (c) Nickel anode and soluble nickel salt [2]. (d)  $\text{Ni}^{2+}$  moves to cathode [1], gains electrons [1], and deposits as Ni [1].

14. 14(a)  $\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$  [2]. (b)  $\text{Cu} \rightarrow \text{Cu}^{2+} + 2\text{e}^-$  [2]. (c) Cathode gains mass; anode loses mass [2]. (d)  $\text{Cu}^{2+}$  removed is replaced by dissolving anode [2].

15. 15(a)  $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$  [2]. (b) Water only at point of use; continuous electricity / quiet [2]. (c) Storage, transport, flammability, low density [any 2]. (d) Making hydrogen may use fossil-fuel electricity and release  $\text{CO}_2$  [2].

### Self-assessment

Circle the first incorrect step and classify it as knowledge, equation or formula, mole ratio, unit conversion, observation, practical method or communication.

Evidence used: Cambridge IGCSE Co-ordinated Sciences 0654 syllabus 2025-2027 and June 2026 papers, mark schemes and Principal Examiner Report supplied by the user. All practice questions are newly written.